This full-text paper was peer-reviewed and accepted to be presented at the IEEE WiSPNET 2017 conference.

Enhancement of Dysarthric Speech for Developing an Effective Speech Therapy Tool

Sreeju Sivaram,[1] C. Santhosh Kumar[2] and A. Anand Kumar[3]
[1,2]Machine Intelligence Research Laboratory
Department of Electronics and Communication Engineering
Amrita School of Engineering, Coimbatore
[3]Department of Neurology, Amrita Institute of Medical Sciences, Kochi
[1,2,3]Amrita Vishwa Vidyapeetham, Amrita University, India-641112
Email: [1]sreejusivaram47@gmail.com [2]cs_kumar@cb.amrita.edu [3]anandkumar@aims.amrita.edu

***Abstract*—Dysarthria is a neuromuscular disorder that results from weakened movement of muscles used in speech production. This results in poor articulation causing the dysarthric speech to be slurred and difficult to understand. The natural auditory feedback makes the patients understand that their speech is of low quality, and this lowers their self confidence and they become more and more introverted causing the disorder to aggravate.**

In this work, we enhance the dysarthric speech and provide the enhanced speech to the patient through auditory feedback. This helps the patients to feel comfortable with their speech and gradually develop confidence to speak more and hence achieve a speedy rehabilitation.

The utterances are analyzed using linear predictive coding (LPC). The LPC features in the acoustic space of dysarthric speaker are mapped to the feature space of the normative population using constrained maximum likelihood linear regression (CMLLR) before they are used for re-synthesising the enhanced speech. We then evaluated the quality of the enhanced speech using subjective and objective measures, DMOS and PESQ, and obtained an improvement of 63% and 43.4% for DMOS and PESQ measures respectively.

Clinical trials are being pursued at Amrita Institute of Medical Sciences, Kochi on patients with dysarthria to evaluate the effectiveness of the proposed approach for faster rehabilitation of the patients. The results of these clinical trials will be reported in due course of time.

***Index Terms*—Dysarthria, feature space mapping, constrained maximum likelihood linear regression, instantaneous auditory feedback, gaussian mixture mapping.**

I. Introduction

We express thoughts and feelings through speech. The process by which thoughts are translated into speech involves the use of articulators for producing various speech sounds. Speech plays a major role in an individuals participation in the society. Various voice and hearing disorders limits the active participation of the person in the societal activities.

Dysarthria is a speech disorder that is caused due to lack of control of various organs that are used in the production of speech. The mechanisms involved in speech production includes respiration, phonation, resonance, articulation and prosody [1]. It affects the speed and strength of the speech signal that is produced. Dysarthric speech exhibits slow rate when compared with normative speech, but certain words within the speech is produced at a faster rate. Further, the rigidity of tongue and lips results in reduced range of movement of speech musculature. The speech sounds tends to be slurred causing the speech to be less intelligible [9].

Dysarthria causes difficulty in coordinating and controlling the muscles used in the production of speech including tongue, lips and vocal folds. It affects the articulation of different types of speech sounds causing the speech to be slurred. Dysarthric speech is characterized by low pitch with pitch breaks occurring in some cases. Patients suffering from dysarthria also have problems controlling the loudness, range, timing, breath, rhythm, and voice qualities of the speech due to lack of proper muscular control. Dysarthria is caused mainly due to neurological disorders like stroke, brain injury, multiple sclerosis and brain tumor which results in weakness or inability to control the articulatory organs. Dysarthric patients have difficulty in communicating because of reduced speech quality and this limits their participation in the society. This led to the introduction of various speech transformation methods to enhance the dysarthric speech intelligibility.

The low quality speech heard by the patients through natural auditory feedback causes the patient to lose confidence to speak more. This causes an inhibition within the patients and they tend to be introverted. Our work aims at developing a speech transformation system that can modify dysarthric speech into highly intelligible speech. The linear predictive parameters in dysarthric speech space are mapped to features in the acoustic space of normative population using constrained maximum likelihood linear regression (CMLLR) with a single frame delay which is not perceptually significant. The modified speech is evaluated using subjective as well as objective measures. The proposed algorithm is used to develop a speech therapy tool that makes use of the transformed speech for providing instantaneous auditory feedback. The tool helps the patients to hear an improved version of their own speech. This in turn can help the patients to speak more and improve their speech intelligibility.

A. Related Work

Many researchers have explored the enhancement of dysarthric speech. Previous studies suggest that auditory

978-1-5090-4442-9/17/$31.00 ©2017 IEEE

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 08:56:39 UTC from IEEE Xplore. Restrictions a

feedback improves the fluency, intelligibility and reduces the rate of the speech. This is effective in the treatment of dysarthria associated with nervous disorders [11]. Existing work in this area based on dynamic time warping algorithm [6] was focused on enhancing dysarthric speech to provide delayed enhanced auditory feedback but it cannot replace the natural auditory feedback which is instantaneous. Frank Rudzicz [1] proposed a method that transformed the dysarthric speech into more intelligible speech. He used acoustic transformations for correcting the pronunciation errors that included phoneme insertion and deletion errors, and adjusting the temporal and frequency characteristics. But his work was not intended to enhance the speech itself.

Kain [2] used a system that transformed the vowels of dysarthric speech so as to match the vowels present in the speech of normative speaker. The transformations were based on analysis of features that improved speech perception. Vijayalakshmi *et al.* [3] used a method that located the wrongly uttered phonemes in dysarthric speech and replaced it with phonemes present in normal speech. This resulted in discontinuities in energy, pitch and formant contour at the concatenation points C1 and C2 making the output speech signal to sound unnatural. The discontinuity in energy was rectified by smoothening energies before and after the concatenation points. Also the pitch and formants in the output speech were adjusted to resemble those present in the normal speech. Divya Das [4] refined the formant trajectories for enhancing the intelligibility of dysarthric speech. In another work, Vijayalakshmi *et al.* [5] made use of speech recognition and synthesis systems to enhance dysarthric speech. The speech recognition system identified the wrongly uttered units. These units were replaced and the speech with higher degree of intelligibility adapted to the patient's voice was generated using the speech synthesis system. Selouni *et al.* [12] proposed a concatenating technique and a grafting method to correct the poorly uttered phonemes.

Previous works are limited to offline enhancement as they require the entire speech utterance before enhancement can take place. The proposed work is based on online enhancement of dysarthric speech as the speech is enhanced as and when it is uttered by the patient with a single frame delay. In this work, the transformed speech is used to develop an effective tool that can be used in speech therapy. The modified speech is fed as instantaneous auditory feedback to replace the natural auditory feedback, provided the feedback signal is of appreciable quality. Any errors introduced by the auditory feedback can deteriorate speech intelligibility [8]. The patient hears the enhanced speech through instantaneous auditory feedback and is convinced that the speech is of good quality. This helps to develop confidence within the patients to speak more and more and to regain natural speech through therapy.

The paper is organized as follows: Section II deals with the methodology being implemented. Section III discusses the results based on objective and subjective measures. Section IV concludes the paper.

II. Methodology

Constrained maximum likelihood linear regression (CMLLR) is a linear feature space transformation technique. It calculates transformation matrix that is used to reduce the mismatch between normal speech and dysarthric speech. Thus CMLLR is a feature adaptation method that computes linear transformations for the features to shift dysarthric speech so as to match the normal speech.

Speech from one acoustic feature space can be transformed to another feature space by performing gaussian mixture mapping [2]. The features in the dysarthric speech feature space are mapped to features in the good speech feature space. This mapping is performed in a linear manner using constrained maximum likelihood linear regression (CMLLR) technique. The relationship between the normative speech y and dysarthric speech x differs depending on the speakers and is not known in advance (apriori). A gaussian mixture model (GMM) is used to model the probability distribution of the source spectral features x as the sum of Q normal distributions with mean vector μ, diagonal covariance matrix σ, and prior probability α and is given by Eq. (1)

$$p(x) = \sum_{q=1}^{Q} \alpha_q N(x; \mu_q, \sigma_q). \tag{1}$$

The GMM is trained in an unsupervised case. In the unsupervised case, EM algorithm is used for estimating the GMM parameters by varying the mixture components [1]. A linear feature space transformation, CMLLR is used to map the syllables uttered by dysarthric and normative speakers. The transformation matrix W and bias vector b are calculated for each syllable using the CMLLR algorithm.

The transformation is then applied to each syllable of the dysarthric speech feature. Thus mismatch between the normative speech feature and dysarthric features are reduced. In this work, LPC features extracted from the utterances of Nemours database [7] are used for performing the transformation. The predicted feature is then re-synthesised using LPC synthesis technique to obtain the transformed speech. The proposed system architecture is as shown in Fig. 1.

III. Results and Discussions

Feature space mapping of LPC parameters is implemented using CMLLR in MATLAB. Significant enhancement of intelligibility of the speech at syllable level is obtained. LPC analysis technique is used to obtain LPC coefficients and LPC residual signal. The transformation using CMLLR is applied to each syllable which results in transformed features. The enhanced speech is then re-synthesised by LPC synthesis technique using the transformed features and residual signal [10]. The LPC analysis and LPC synthesis techniques are performed with the help of auditory toolbox. The transformed features re-synthesised using the residual of the dysarthric speaker retained the speaker identity. But speaker identity is lost when the speech is re-synthesised using the residual of normal

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 08:56:39 UTC from IEEE Xplore. Restrictions a

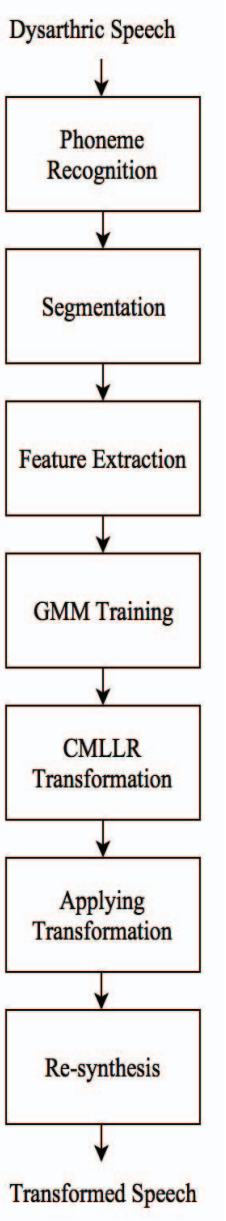


Fig. 1. System architecture for dysarthric speech enhancement using CMLLR.

speaker. The best performance was obtained for a mixture of 8 gaussians.

The enhanced speech quality is evaluated using subjective as well as objective scores. The objective scores of the transformed speech improved significantly. The quality of the speech is assessed before and after enhancement objectively using perceptual evaluation of speech quality (PESQ) measure. The intelligibility scores of preliminary subjective test using degradation mean opinion score (DMOS) also improved.

The DMOS scores of the transformed syllables are measured and is given in Tables I through III. DMOS is a listener based test which is used to evaluate the perceptual quality of the enhanced speech signal. The listeners rate the speech

TABLE I
DMOS SCORES FOR ENHANCED SYLLABLE /AA/ RE-SYNTHESIZED USING NORMAL AND DYSARTHRIC RESIDUAL SIGNALS.

Using residual of normal speaker	4.30	4.25	4.29	4.35	4.40	4.20	4.35
Using residual of dysarthric speaker	2.50	2.32	2.35	3.00	2.50	2.45	2.69

TABLE II
DMOS SCORES FOR ENHANCED SYLLABLE /AE/ RE-SYNTHESIZED USING NORMAL AND DYSARTHRIC RESIDUAL SIGNALS.

Using residual of normal speaker	4.00	4.15	4.10	4.25	4.33	4.25	4.15
Using residual of dysarthric speaker	2.30	2.48	2.35	2.55	2.32	3.00	2.29

TABLE III
DMOS SCORES FOR ENHANCED SYLLABLE /N/ RE-SYNTHESIZED USING NORMAL AND DYSARTHRIC RESIDUAL SIGNALS.

Using residual of normal speaker	4.20	4.28	4.23	4.31	4.26	4.30	4.15
Using residual of dysarthric speaker	2.30	2.20	2.45	2.35	2.30	2.56	2.90

TABLE IV
PESQ SCORES FOR SYLLABLES BEFORE AND AFTER ENHANCEMENT.

Syllable	Before enhancement	After enhancement
/aa/	1.25	3.42
/ae/	1.34	3.21
/n/	1.02	3.18

intelligibility on a scale of 1–5. The perceptual quality of the speech was also assessed objectively using PESQ. P.862 is an International Telecommunication Unit (ITU) recommendation for evaluation of the speech quality. It presents an easy and objective method for the evaluation of the transformed speech. PESQ returns a score from 1 to 5 (5-excellent 4-good 3-average 2-poor 1-bad). PESQ scores obtained for particular syllables /aa/ /ae/ and /n/ before and after enhancement using CMLLR technique is given in Table IV.

IV. CONCLUSION

In this work, the intelligibility of dysarthric speech is improved using feature space mapping of LPC features towards developing an effective therapy tool. Constrained maximum likelihood linear regression (CMLLR) technique is used to map the features in the acoustic space of dysarthric speaker to the features in the normative feature space. The transformation was carried out on syllables of utterances of dysarthric patients from the Nemours database. The intelligibility of the modified speech is measured subjectively using DMOS listening test and objectively using PESQ. We have observed significant enhancements over the dysarthric speech with an intelligibility improvement of 63% and 43.4% for DMOS and PESQ measures respectively. This approach is used in building a therapy tool which is simple, robust, portable and

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 08:56:39 UTC from IEEE Xplore. Restrictions a

cost effective. The tool makes use of the enhanced speech to provide instantaneous auditory feedback. The therapy tool therefore supports the patients to regain self confidence to interact more thereby improving their participation in the society.

Clinical trials are being pursued at Amrita Institute of Medical Sciences to evaluate the efficiency of the proposed approach. The results of clinical trials will be reported in due course of time.

Acknowledgements: The authors would like to thank Sreekumar K.T, Kuruvachan K George, Arundas K, S. Arun Kumar of Machine Intelligence Research Laboratory, Amrita School of Engineering, and the participants of the listening test for their support and suggestions that helped in the successful completion of the work.

REFERENCES

[1] F. Rudzicz, "Adjusting dysarthric speech signals to be more intelligible," *Computer Speech and Language*, vol. 27, no. 6, pp. 1163–1177, 2013, special Issue on Speech and Language Processing for Assistive Technology.

[2] A. B. Kain, J. P. Hosom, X. Niu, J. P. Van Santen, M. Fried-Oken, and J. Staehely, "Improving the intelligibility of dysarthric speech," in *Speech Communication*, vol. 49, no. 9, pp. 743–759, 2007.

[3] M. P. Vijayalakshmi Saranya and Nagarajan Thangavelu, "Improving the intelligibility of dysarthric speech by modifying system parameters, retaining speaker's identity." in *Recent Trends In Information Technology (ICRTIT), 2012 International Conference on*, pp. 60–65. IEEE, 2012.

[4] Divya Das, C. Santhosh Kumar, and P. C. Reghu Raj, "Dysarthric speech enhancement using formant trajectory refinement," *International Journal of Latest Trends in Engineering and Technology (IJLTET)*, vol. 2 no. 4 July 2013.

[5] M. Dhanalakshmi and P. Vijayalakshmi, "Intelligibility modification of dysarthric speech using HMM-based adaptive synthesis system," in *Biomedical Engineering (ICoBE), 2015 2nd International Conference on*. IEEE, 2015.

[6] S. Arun Kumar and C. Santhosh Kumar, "Improving the intelligibility of dysarthric speech towards enhancing the effectiveness of speech therapy." in *Advances in Computing, Communications and Informatics (ICACCI), 2016 International Conference on*, pp. 1000–1005. IEEE, 2016.

[7] Xavier Menendez-Pidal, James B. Polikoff, Shirley M. Peters, Jennie E. in Leonzio, and H. T. Bunnel, "The nemours database of dysarthric speech," in *Spoken Language, 1996. ICSLP 96. Proceedings., Fourth International Conference on*., vol. 3, pp. 1962–1965.

[8] Spille, Constantin, Hendrik Kayser, Hynek Hermansky, and Bernd T. Meyer, "Assessing speech quality in speech-aware hearing aids based on phoneme posteriorgrams," *Interspeech 2016*, 2016, pp. 1755–1759.

[9] Joseph R. Duffy, "Motor speech disorders: substrates," *Differential Diagnosis, and Management*. Elsevier Health Sciences, 2013.

[10] Lawrence R. Rabiner and Ronald W. Schafer, *Digital processing of speech signals*, Prentice Hall, 1978.

[11] P. G Blanchet and P. R Hoffman, "Factors influencing the effects of delayed auditory feedback on dysarthric speech associated with parkinsons disease," in *Journal of Communication Disorders, Deaf Studies Hearing Aids*, 2014.

[12] Mohammed Sidi Yakcoub, Sid-Ahmed Selouani, and Douglas O'shaughnessy, "Speech assistive technology to improve the interaction of dysarthric speakers with machines," in *International Symposium on Communications, Control and Signal Processing (ISCCSP) 2008*, pp. 1150–1154.

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 08:56:39 UTC from IEEE Xplore. Restrictions a